Sadhasivam Perichi

Email: sadhasivamperichi@gmail.com
LinkedIn: linkedin.com/sadhasivam-perichi-160449255/
Mobile: +91-9361215021
Github: github.com/smily-sadha
Portfolio: sadha-portfolio.netlify.app

EDUCATION

- **Sona College of Technology** — Salem, India
  Bachelor of Engineering - Artificial Intelligence & Machine Learning CGPA: 7.5 — *2022 – 2026*
- **IVL Matric HR. Sec. School** — Dharmapuri, India
  HSC; 81.5% — *2021 – 2022*
- **IVL Matric HR. Sec. School** — Dharmapuri, India
  SSLC; 93.5% — *2019 – 2020*

INTERNSHIP EXPERIENCE

- **Spacemarvel.ai – AI Engineer Intern** — *July 2025 – Present*
  Designed and shipped production-grade agentic voice solutions for real-world call workflows including cold-calling automation, hospital appointment booking, delivery coordination, and AI-driven lead generation. Built a Retrieval-Augmented Generation (RAG) system for hospital SOP assistance using LangChain, implementing document ingestion, chunking, embedding, and retrieval pipelines for knowledge-grounded responses. Engineered real-time speech pipelines integrating Deepgram STT, LLM reasoning, ElevenLabs TTS, and WebRTC transport, achieving sub-second conversational turnarounds depending on network and model load. Productionized end-to-end STT → LLM → TTS automation, reducing manual call-handling effort by approximately 60% and enabling scalable AI voice-agent deployments. Integrated Pipecat and LiveKit for low-latency conversational orchestration and implemented authentication/session management using Clerk.

PROJECTS

- **Online Voting System with Face Recognition:**
  Engineered a secure online voting platform using facial biometrics for real-time voter authentication. Implemented OpenCV preprocessing and Single Shot Detection for robust face localization. Built identity-validation logic to prevent duplicate voting and ensured secure vote validation and persistence using MongoDB, along with interactive voter registration and ballot submission workflows.
  Tech Stack: Python, OpenCV, Single Shot Detection, MongoDB
- **Plant Leaf Disease Detection System:**
  Built a production-grade CNN classifier for plant leaf disease detection with preprocessing, augmentation, and normalization pipelines for improved generalization across multiple plant species. Designed TensorFlow/Keras training workflows with cross-validation, learning-rate scheduling, and hyperparameter tuning to optimize accuracy and recall. Deployed the model in a Flask web application with optimized inference pipelines, achieving inference latency below 300 ms on GPU and approximately 1.2 s on CPU, while achieving approximately 92% top-1 accuracy on benchmark datasets.
  Tech Stack: Python, TensorFlow/Keras, OpenCV, Flask
- **Online Job Application Web Portal:**
  Developed a full-stack job application platform supporting secure authentication, role-based access control, and end-to-end recruiter/applicant workflows. Implemented RESTful APIs using Node.js and Express with Clerk authentication, scalable job listing endpoints, and optimized MongoDB schemas with indexing and rate limiting. Built a responsive React frontend with form validation, resumable file uploads, and real-time application status notifications.
  Tech Stack: React, Node.js, Express, MongoDB, Clerk
- **Online Attendance System using Face Recognition:**
  Automated attendance capture using live webcam streams with CNN-based feature extraction and lightweight SVM classifiers for efficient face recognition. Engineered preprocessing pipelines including

CLAHE, histogram equalization, facial alignment, and temporal smoothing to improve recognition stability under varying lighting conditions. Optimized frame processing using ROI tracking and asynchronous capture, achieving greater than 95% attendance accuracy during controlled indoor evaluations.
Tech Stack: Python, OpenCV, CNN, SVM

- **FINAL YEAR PROJECT – Real-Time Object Detection & Audio Guidance System for the Visually Impaired:**
  Engineered an end-to-end assistive system combining real-time object detection (YOLOv8), face recognition (InsightFace embeddings + FAISS), and audio feedback (Deepgram TTS) into a low-latency pipeline running at approximately 2–3 FPS on CPU hardware. Designed a dual-track architecture with parallel vision modules (YOLOv8, SSD face detection, face embedding, emotion recognition) and audio modules (Deepgram STT, voice activity detection) feeding a central decision unit for distance estimation and context-aware response generation using Gemini LLM. Implemented dynamic face enrollment with voice-driven registration, multi-angle image capture, InsightFace embedding generation, and real-time FAISS index updates without system restarts. Built a wearable hardware prototype using ESP32-CAM, Arduino Nano, and HC-SR04 ultrasonic sensors to combine vision-based detection with sensor-based obstacle ranging for multimodal environmental awareness. Achieved person detection accuracy above 0.90 YOLO confidence and reliable named face recognition across varied indoor lighting conditions, delivering low-latency contextual audio guidance. Presented the work at ICCISS 2026: *"IoT-Based Smart Assistive Vision System For Blind Navigation Using Deep Learning."*
  Tech Stack: YOLOv8, InsightFace, FAISS, Deepgram, OpenCV, Python, Gemini LLM, ESP32-CAM, Arduino

SKILLS SUMMARY

- **Programming:** Java, Python
- **AI / ML / DL:** PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, CNN, Transfer Learning
- **LLM & Agentic AI:** LangChain, LangGraph, ChromaDB, RAG, OpenAI API, Pipecat, Deepgram
- **Web & Backend:** React, Node.js, Express, MongoDB Atlas, MySQL, Clerk
- **Data Analytics:** NumPy, Pandas, Matplotlib, Seaborn, Excel, Power BI, Tableau

AREA OF INTEREST

- AI Engineer
- Data Analytics
- Full-Stack Web Development

CERTIFICATION

- NoviTech – Data Analyst, Artificial Intelligence, Machine Learning, Full Stack Developer
- Coursera – Data Structures
- Infosys – Foundation of Java
- Guvi – Basics of Python
- NPTEL – Design & Implementation of Human Computer Interface

ACHIEVEMENTS

- Secured **1st place** in Football (2022), Sona College of Technology
- Secured **3rd place** in UiPath Hackathon (2024), Sona College of Technology
- secured **3rd place** in paper presentation (2024) , sona college of technology